IE 7615: Neural Networks/Deep Learning

Generative Project Milestone – 1

GROUP 6

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GITHUB LINK : https://github.com/poornikajalavadi/Generative-Project-Neural-Networks?tab=readme-ov-file#1-setup-and-dependencies

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ABSTRACT

This project focuses on building a data preprocessing pipeline for an image captioning system by combining computer vision and natural language processing techniques. A subset of the COCO dataset was used, and image-caption pairs were prepared after handling missing data. Images were converted into 512-dimensional embeddings using the CLIP model, while captions were tokenized using the GPT-2 tokenizer with custom special tokens. The pipeline was validated through similarity checks and visualization outputs. The processed data was saved for future use in model training. This milestone establishes a strong foundation for developing an image caption generation model in the next stage.

INTRODUCTION

In recent years, combining computer vision and natural language processing has become an important area in artificial intelligence. One of the most practical applications of this combination is image captioning, where a model generates meaningful textual descriptions for images. This has real-world applications in accessibility systems for visually impaired users, image search engines, and content organization.

In this project, the objective of Milestone 1 is to build a strong data processing pipeline for an image captioning system. Instead of directly training a model, the focus is on preparing both image and text data in a structured and usable format. This includes extracting meaningful features from images and converting captions into numerical representations that can be used in later stages.

ENVIRONMENT SETUP AND CONFIGURATION

The implementation was carried out using Python in a Google Colab environment with GPU support (Tesla T4). The following libraries were used: PyTorch for deep learning operations, Hugging Face Transformers for CLIP and GPT-2 models, PyCOCOTools for handling COCO dataset annotations, and PIL along with Matplotlib for image processing and visualization.

To ensure reproducibility and consistency across runs, the following configuration settings were used. A fixed random seed of 42 was applied. A subset of 5000 images was selected from the dataset. The maximum caption length was set to 64 tokens. The computations were performed using a CUDA-enabled GPU, which helped in faster processing of image embeddings.

DATASET DESCRIPTION

The COCO (Common Objects in Context) dataset was used for this project, as it is one of the most widely used datasets for image captioning tasks. It contains a large number of images, each paired with multiple human-written captions.

A subset of 5000 images was randomly selected from the training dataset to make the computation manageable. During preprocessing, several images were found to be missing or inaccessible. Because of this, the final dataset consisted of 2811 valid image-caption pairs.

Each data point in the dataset includes an image file path and its corresponding caption. This step highlighted a real-world issue where datasets are often imperfect and require cleaning before use. By removing invalid entries, the quality of the dataset was improved.

METHODOLOGY / PIPELINE

The pipeline for this milestone is divided into three main components: data preparation, image processing, and text processing.

Data Preparation

The dataset was downloaded and stored in Google Drive. Image files and annotation files were extracted and organized into appropriate directories. For each selected image, one caption was chosen and paired with it to form a structured dataset.During this step, missing or corrupted image files were skipped. Although this reduced the total number of usable samples, it ensured that the dataset remained clean and reliable.

Image Processing using CLIP

To extract meaningful visual features from images, the CLIP (Contrastive Language–Image Pretraining) model was used. Each image was converted into a 512-dimensional embedding vector, which captures the semantic meaning of the image.

The images were processed in batches to improve efficiency. After generating embeddings, they were normalized so that cosine similarity could be used for comparison.

Sanity checks were performed to validate the embeddings. When an image was compared with itself, the similarity score was close to 1. When different images were compared, the similarity scores were lower. This confirmed that the embeddings were correctly generated.

In addition, a similarity search experiment was conducted. For a given image, the most similar images were retrieved using cosine similarity. The results showed that visually similar images were grouped together, indicating that the model successfully captured meaningful visual features.

Text Processing using GPT-2 Tokenizer

The captions were processed using the GPT-2 tokenizer. Since GPT-2 does not include certain special tokens by default, custom tokens were added, including a start token, an end token, and a padding token.

Each caption was converted into a sequence of numerical token IDs. All captions were padded or truncated to a fixed length of 64 tokens to maintain consistency.

Attention masks were also generated to distinguish between actual text and padding tokens. This ensures that the model can correctly interpret the input during training.

RESULTS AND OBSERVATIONS

Several important observations were made during this milestone. Although 5000 images were initially selected, only 2811 valid image-caption pairs were obtained due to missing or inaccessible files. This highlights the importance of data cleaning in real-world projects.

The CLIP model successfully generated meaningful 512-dimensional embeddings that captured visual similarities between images. Sanity checks confirmed that the embeddings were properly normalized and reliable.

The similarity search results showed that visually related images were correctly identified. Caption tokenization was efficient, with most captions being relatively short and no truncation issues observed.

Overall, the pipeline successfully processed both image and text data and prepared them for the next stage of model development.

SAMPLE OUTPUT

Figure 1: Sample image-caption pairs from the dataset. Each image is paired with a descriptive caption, showing that the dataset has been correctly processed and aligned.

Figure 2: CLIP embedding similarity search for a food-related image. The model successfully retrieves visually similar images containing dishes and food arrangements, demonstrating that the embeddings capture meaningful visual patterns.

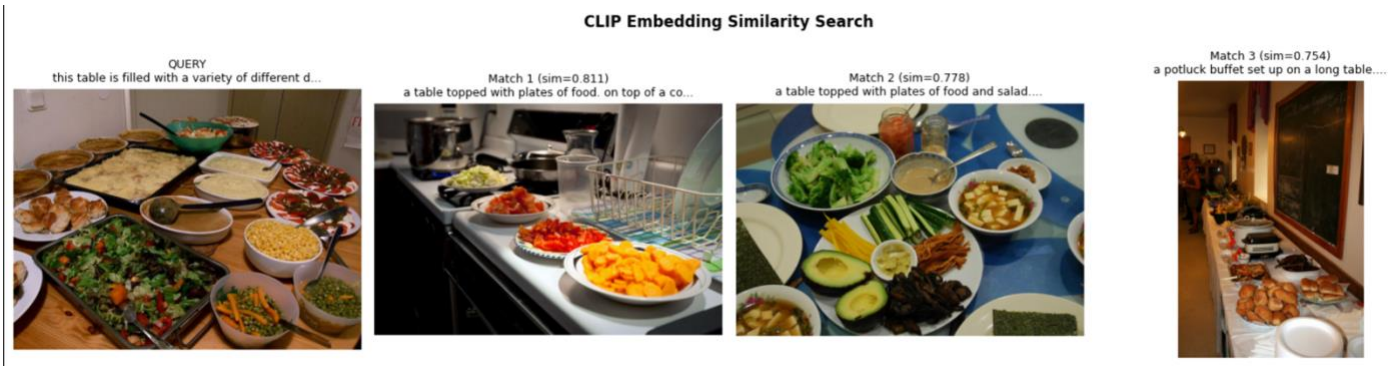

Figure 3: CLIP embedding similarity search for a hotel room image. The retrieved results show similar indoor scenes such as bedrooms and furnished spaces, indicating that the model effectively understands scene-level features.

GENERATED FILES

At the end of the pipeline, all processed data was saved for reuse. The following files were generated:

- milestone1_tensors.pt
  Contains image embeddings (512-dimensional), tokenized captions, and attention masks
- milestone1_metadata.json
  Contains dataset information such as image paths, captions, and configuration details
- Tokenizer folder
  Stores the GPT-2 tokenizer with added special tokens

These files allow the processed data to be easily loaded in future stages without repeating preprocessing.

GITHUB REPOSITORY

The complete project has been organized and uploaded to a GitHub repository for version control and reproducibility.

GitHub Repository: https://github.com/poornikajalavadi/Generative-Project-Neural-Networks?tab=readme-ov-file#1-setup-and-dependencies

The repository includes the implementation of the Milestone 1 pipeline along with proper documentation. It contains the Jupyter Notebook used for data preprocessing, including dataset preparation, CLIP-based image embedding generation, and GPT-2 caption tokenization.

The repository is structured to clearly separate different stages of the pipeline, including environment setup, dataset processing, embedding generation, and saving processed outputs. It also includes explanations of key steps such as dependency setup, data handling, and validation through sanity checks.

This organization ensures that the project is easy to understand, reproduce, and extend for future milestones.

CONCLUSION

In this milestone, a complete data preprocessing pipeline for an image captioning system was successfully developed. Images were converted into meaningful embeddings using CLIP, and captions were transformed into structured token sequences using the GPT-2 tokenizer.

Despite challenges such as missing data, a clean and usable dataset was prepared. The processed outputs have been saved efficiently and are ready to be used for training a caption generation model in the next phase.

This milestone provides a strong foundation for building a complete image captioning system in future work.